NOP (00)	Cycles: 2	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2		
2.1		
2.2	Read **PC**	Store as OpCode.
+X.1		

TCALL (01)	Cycles: 8	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Operand.
2.1		
2.2		
3.1		
3.2	Write to u8(**SP**+0) | $0100	Push **PC**.H onto stack.
4.1		
4.2	Write to u8(**SP**-1) | $0100	Push **PC**.L onto stack.
5.1		Dec. **SP** by 2.
5.2		
6.1		
6.2	Read $FFDE	Store as **PC**.L.
7.1		
7.2	Read $FFDF	Store as **PC**.H.
8.1		
8.2	Read **PC**	Store as OpCode.
+X.1		

SET1 (02)	Cycles: 4	Size: 2
	Direct Page Bit d.b (Read/Modify/Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Operand.
2.1		Inc. **PC**. Address = (Operand | ((**PSW** & **P** flag) << 3)).
2.2	Read Address	Store as Operand.
3.1		Operand = (Operand & ~$01) | $01.
3.2	Write to Address	Write Operand.
4.1		
4.2	Read **PC**	Store as OpCode.
+X.1		

BBS0 (03)	Cycles: 5-7	Size: 3
	Direct Page Bit Relative d.b, r	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Operand.
2.1		Inc. **PC**. Address = (Operand | ((**PSW** & **P** flag) << 3)).
2.2	Read Address	Store as Operand.
3.1		Set Jump to (Operand & (1 << 0)) == (1 << 0).
3.2		
4.1		
4.2	Read **PC**	Store as Operand.
5.1		Inc. **PC**. If !Jump, end instruction (next half-cycle is 7.2).
5.2		
6.1		Address = **PC** + i8(Operand). **PC**.L = Address.L.
6.2		
7.1		**PC**.H = Address.H.
7.2	Read **PC**	Store as OpCode.
+X.1		

OR (04)	Cycles: 3	Size: 2
	Direct Page d (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Operand.
2.1		Inc. **PC**. Address = (Operand | ((**PSW** & P flag) << 3)).
2.2	Read Address	Store as Operand.
3.1		**A** |= Operand. **N** flag = (**A** & $80), **Z** flag = !**A**.
3.2	Read **PC**	Store as OpCode.
+X.1		

OR (05)	Cycles: 4	Size: 3
	Absolute !a (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Address.L.
2.1		Inc. **PC**.
2.2	Read **PC**	Store as Address.H.
3.1		Inc. **PC**.
3.2	Read Address	Store as Operand.
4.1		**A** |= Operand. **N** flag = (**A** & $80), **Z** flag = !**A**.
4.2	Read **PC**	Store as OpCode.
+X.1		

OR (06)	Cycles: 3	Size: 1
	Indirect X (X) (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2		
2.1		
2.2	Read (**X** | ((**PSW** & P flag) << 3))	Store as Operand.
3.1		**A** |= Operand. **N** flag = (**A** & $80), **Z** flag = !**A**.
3.2	Read **PC**	Store as OpCode.
+X.1		

OR (07)	Cycles: 6	Size: 2
	X-Indexed Indirect [d+X] (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Operand.
2.1		Inc. **PC**.
2.2		
3.1		Address = ((**X** + Operand) & $FF) | ((**PSW** & P flag) << 3).
3.2	Read Address	Store as Pointer.L.
4.1		
4.2	Read ((Address + 1) & $FF) | ((**PSW** & **P** flag) << 3)	Store as Pointer.H.
5.1		
5.2	Read Pointer	Store as Operand.
6.1		**A** |= Operand. **N** flag = (**A** & $80), **Z** flag = !**A**.
6.2	Read **PC**	Store as OpCode.
+X.1		

OR (08)	Cycles: 2	Size: 2
	Immediate	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Operand.
2.1		Inc. **PC**. **A** |= Operand. **N** flag = (**A** & $80), **Z** flag = !**A**.
2.2	Read **PC**	Store as OpCode.
+X.1		

OR (09)	Cycles: 6	Size: 3
	Direct Page to Direct Page dd, ds (Read/Modify/Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Operand.
2.1		Inc. **PC**. Address = (Operand | ((**PSW** & **P** flag) << 3)).
2.2	Read Address	Store as Operand0.
3.1		
3.2	Read **PC**	Store as Operand.
4.1		Inc. **PC**. Address = (Operand | ((**PSW** & **P** flag) << 3)).
4.2	Read Address	Store as Operand1.
5.1		Operand0 |= Operand1. **N** flag = (Operand0 & $80), **Z** flag = !Operand0.
5.2	Write to Address	Write Operand0.
6.1		
6.2	Read **PC**	Store as OpCode.
+X.1		

OR1 (0A)	Cycles: 5	Size: 3
	Absolute Boolean Bit m.b	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Address.L.
2.1		Inc. **PC**.
2.2	Read **PC**	Store as Address.H.
3.1		Inc. **PC**.
3.2	Read (Address & $1FFF)	Store as Operand.
4.1		Bit = Address >> 13.
		C flag |= ((Operand & (1 << Bit)) != 0).
4.2		
5.1		
5.2	Read **PC**	Store as OpCode.
+X.1		

ASL (0B)	Cycles: 4	Size: 2
	Direct Page d (Read/Modify/Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Operand.
2.1		Inc. **PC**. Address = (Operand | ((**PSW** & P flag) << 3)).
2.2	Read Address	Store as Operand.
3.1		**C** flag = (Operand & $80). Operand <<= 1. **N** flag = (Operand & $80), **Z** flag = !Operand.
3.2	Write to Address	Write Operand.
4.1		
4.2	Read **PC**	Store as OpCode.
+X.1		

ASL (0C)	Cycles: 5	Size: 3
	Absolute !a (Read/Modify/Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Address.L.
2.1		Inc. **PC**.
2.2	Read **PC**	Store as Address.H.
3.1		Inc. **PC**.
3.2	Read Address	Store as Operand.
4.1		**C** flag = (Operand & $80). Operand <<= 1. **N** flag = (Operand & $80), **Z** flag = !Operand.
4.2	Write to Address	Write Operand.
5.1		
5.2	Read **PC**	Store as OpCode.
+X.1		

PUSH (0D)	Cycles: 4	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Discard.
2.1		
2.2	Write to u8(**SP**+0) | $0100	Push **PSW** onto stack.
3.1		Dec. **SP** by 1.
3.2		
4.1		
4.2	Read **PC**	Store as OpCode.
+X.1		

TSET1 (0E)	Cycles: 6	Size: 3
	Absolute !a (Read/Modify/Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Address.L.
2.1		Inc. **PC**.
2.2	Read **PC**	Store as Address.H.
3.1		Inc. **PC**.
3.2	Read Address	Store as Operand.
4.1		Tmp = (**A** - Operand).
		Operand |= **A**.
		Z flag = (Tmp == 0), **N** flag = (Tmp & $80).
4.2	Read Address	Discard.
5.1		
5.2	Write to Address	Write Operand.
6.1		
6.2	Read **PC**	Store as OpCode.
+X.1		

BRK (0F)	Cycles: 8	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Discard.
2.1		
2.2	Write to u8(**SP**+0) | $0100	Push **PC**.H onto stack.
3.1		
3.2	Write to u8(**SP**-1) | $0100	Push **PC**.L onto stack.
4.1		
4.2	Write to u8(**SP**-2) | $0100	Push **PSW** onto stack.
5.1		Dec. **SP** by 3. **I** flag = 0, **X** flag = 1.
5.2		
6.1		
6.2	Read $FFDE	Store as **PC**.L.
7.1		
7.2	Read $FFDF	Store as **PC**.H.
8.1		
8.2	Read **PC**	Store as OpCode.
+X.1		

BPL (10)	Cycles: 2-4	Size: 2
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**. Set Jump to (**PSW** & $80) == $00.
1.2	Read **PC**	Store as Operand.
2.1		Inc. **PC**. If !Jump, end instruction (next half-cycle is 4.2).
2.2		
3.1		Address = **PC** + i8(Operand). **PC**.L = Address.L.
3.2		
4.1		**PC**.H = Address.H.
4.2	Read **PC**	Store as OpCode.
+X.1		

TCALL1 (11)	Cycles: 8	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Operand.
2.1		
2.2		
3.1		
3.2	Write to u8(**SP**+0) | $0100	Push **PC**.H onto stack.
4.1		
4.2	Write to u8(**SP**-1) | $0100	Push **PC**.L onto stack.
5.1		Dec. **SP** by 2.
5.2		
6.1		
6.2	Read $FFDC	Store as **PC**.L.
7.1		
7.2	Read $FFDD	Store as **PC**.H.
8.1		
8.2	Read **PC**	Store as OpCode.
+X.1		

CLR1 (12)	Cycles: 4	Size: 2
	Direct Page Bit d.b (Read/Modify/Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Operand.
2.1		Inc. **PC**. Address = (Operand | ((**PSW** & **P** flag) << 3)).
2.2	Read Address	Store as Operand.
3.1		Operand = (Operand & ~$01) | $00.
3.2	Write to Address	Write Operand.
4.1		
4.2	Read **PC**	Store as OpCode.
+X.1		

BBC0 (13)	Cycles: 5-7	Size: 3
	Direct Page Bit Relative d.b, r	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Operand.
2.1		Inc. **PC**. Address = (Operand | ((**PSW** & **P** flag) << 3)).
2.2	Read Address	Store as Operand.
3.1		Set Jump to (Operand & (1 << 0)) == (0 << 0).
3.2		
4.1		
4.2	Read **PC**	Store as Operand.
5.1		Inc. **PC**. If !Jump, end instruction (next half-cycle is 7.2).
5.2		
6.1		Address = **PC** + i8(Operand). **PC**.L = Address.L.
6.2		
7.1		**PC**.H = Address.H.
7.2	Read **PC**	Store as OpCode.
+X.1		

OR (14)	Cycles: 4	Size: 2
	X-Indexed Direct Page d+X (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Operand.
2.1		Inc. **PC**.
2.2		
3.1		Address = ((**X** + Operand) & $FF) | ((**PSW** & **P** flag) << 3).
3.2	Read Address	Store as Operand.
4.1		**A** |= Operand. **N** flag = (**A** & $80), **Z** flag = !**A**.
4.2	Read **PC**	Store as OpCode.
+X.1		

OR (15)	Cycles: 5	Size: 3
	X-Indexed Absolute !a+X (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Address.L.
2.1		Inc. **PC**.
2.2	Read **PC**	Store as Address.H.
3.1		Inc. **PC**.
3.2		
4.1		Pointer = (**X** + Address).
4.2	Read Pointer	Store as Operand.
5.1		**A** |= Operand. **N** flag = (**A** & $80), **Z** flag = !**A**.
5.2	Read **PC**	Store as OpCode.
+X.1		

OR (16)	Cycles: 5	Size: 3
	Y-Indexed Absolute !a+Y (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Address.L.
2.1		Inc. **PC**.
2.2	Read **PC**	Store as Address.H.
3.1		Inc. **PC**.
3.2		
4.1		Pointer = (**Y** + Address).
4.2	Read Pointer	Store as Operand.
5.1		**A** |= Operand. **N** flag = (**A** & $80), **Z** flag = !**A**.
5.2	Read **PC**	Store as OpCode.
+X.1		

OR (17)	Cycles: 6	Size: 2
	Indirect Y-Indexed [d]+Y (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Operand.
2.1		Inc. **PC**.
2.2		
3.1		Address = (Operand | ((**PSW** & **P** flag) << 3)).
3.2	Read Address	Store as Pointer.L.
4.1		
4.2	Read ((Address + 1) & $FF) | ((**PSW** & **P** flag) << 3)	Store as Pointer.H.
5.1		Address = (**Y** + Pointer).
5.2	Read Address	Store as Operand.
6.1		**A** |= Operand. **N** flag = (**A** & $80), **Z** flag = !**A**.
6.2	Read **PC**	Store as OpCode.
+X.1		

OR (18)	Cycles: 5	Size: 3
	Immediate Data to Direct Page d, #i (Read/Modify/Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Operand0.
2.1		Inc. **PC**.
2.2	Read **PC**	Store as Operand.
3.1		Inc. **PC**. Address = (Operand | ((**PSW** & **P** flag) << 3)).
3.2	Read Address	Store as Operand1.
4.1		Operand0 |= Operand1. **N** flag = (Operand0 & $80), **Z** flag = !Operand0.
4.2	Write to Address	Write Operand0.
5.1		
5.2	Read **PC**	Store as OpCode.
+X.1		

OR (19)	Cycles: 5	Size: 1
	Indirect Page to Indirect Page	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Discard.
2.1		
2.2	Read (**Y** | ((**PSW** & **P** flag) << 3))	Store as Operand0.
3.1		
3.2	Read (**X** | ((**PSW** & **P** flag) << 3))	Store as Operand1.
4.1		Operand0 |= Operand1. **N** flag = (Operand0 & $80), **Z** flag = !Operand0.
4.2	Write to (**X** | ((**PSW** & **P** flag) << 3))	Write Operand0.
5.1		
5.2	Read **PC**	Store as OpCode.
+X.1		

DECW (1A)	Cycles: 6	Size: 2
	Direct Page d (Read/Modify/Write 16)	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Operand.
2.1		Inc. **PC**. Address = (Operand | ((**PSW** & **P** flag) << 3)).
2.2	Read Address	Store as Operand.
3.1		Underflow = (Operand == 0). Operand -= 1. LowZero = (Operand == 0).
3.2	Write to Address	Write Operand.
4.1		
4.2	Read ((Address + 1) & $FF) | ((**PSW** & **P** flag) << 3)	Store as Operand.
5.1		Operand -= Underflow. **N** flag = (Operand & $80), **Z** flag = (LowZero && !Operand).
5.2	Write to ((Address + 1) & $FF) | ((**PSW** & **P** flag) << 3)	Write Operand.
6.1		
6.2	Read **PC**	Store as OpCode.
+X.1		

ASL (1B)	Cycles: 5	Size: 2
	X-Indexed Direct Page d+X (Read/Modify/Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Operand.
2.1		Inc. **PC**.
2.2		
3.1		Address = ((**X** + Operand) & $FF) | ((**PSW** & **P** flag) << 3).
3.2	Read Address	Store as Operand.
4.1		**C** flag = (Operand & $80). Operand <<= 1. **N** flag = (Operand & $80), **Z** flag = !Operand.
4.2	Write to Address	Write Operand.
5.1		
5.2	Read **PC**	Store as OpCode.
+X.1		

ASL (1C)	Cycles: 2	Size: 1
	Accumulator A	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Discard.
2.1		**C** flag = (**A** & $80). **A** <<= 1. **N** flag = (**A** & $80), **Z** flag = !**A**.
2.2	Read **PC**	Store as OpCode.
+X.1		

DEC (1D)	Cycles: 2	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2		
2.1		**X** -= 1. **N** flag = (**X** & $80), **Z** flag = !**X**.
2.2	Read **PC**	Store as OpCode.
+X.1		

CMP (1E)	Cycles: 4	Size: 3
	Absolute !a (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Address.L.
2.1		Inc. **PC**.
2.2	Read **PC**	Store as Address.H.
3.1		Inc. **PC**.
3.2	Read Address	Store as Operand.
4.1		Tmp = **X** - Operand. **C** flag = (Tmp >= 0), **N** flag = (Tmp & $80), **Z** = !Tmp.
4.2	Read **PC**	Store as OpCode.
+X.1		

JMP (1F)	Cycles: 6	Size: 3
	Absolute X-Indexed Indirect [!a+X]	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Address.L.
2.1		Inc. **PC**.
2.2	Read **PC**	Store as Address.H.
3.1		
3.2		
4.1		Pointer = (**X** + Address).
4.2	Read Pointer	Store as **PC**.L.
5.1		
5.2	Read Pointer + 1	Store as **PC**.H.
6.1		
6.2	Read **PC**	Store as OpCode.
+X.1		

CLRP (20)	Cycles: 2	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2		
2.1		**PSW** = (**PSW** & ~$20).
2.2	Read **PC**	Store as OpCode.
+X.1		

TCALL1 (21)	Cycles: 8	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Operand.
2.1		
2.2		
3.1		
3.2	Write to u8(**SP**+0) | $0100	Push **PC**.H onto stack.
4.1		
4.2	Write to u8(**SP**-1) | $0100	Push **PC**.L onto stack.
5.1		Dec. **SP** by 2.
5.2		
6.1		
6.2	Read $FFDA	Store as **PC**.L.
7.1		
7.2	Read $FFDB	Store as **PC**.H.
8.1		
8.2	Read **PC**	Store as OpCode.
+X.1		

SET1 (22)	Cycles: 4	Size: 2
	Direct Page Bit d.b (Read/Modify/Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Operand.
2.1		Inc. **PC**. Address = (Operand | ((**PSW** & **P** flag) << 3)).
2.2	Read Address	Store as Operand.
3.1		Operand = (Operand & ~$02) | $02.
3.2	Write to Address	Write Operand.
4.1		
4.2	Read **PC**	Store as OpCode.
+X.1		

BBS1 (23)	Cycles: 5-7	Size: 3
	Direct Page Bit Relative d.b, r	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Operand.
2.1		Inc. **PC**. Address = (Operand | ((**PSW** & **P** flag) << 3)).
2.2	Read Address	Store as Operand.
3.1		Set Jump to (Operand & (1 << 1)) == (1 << 1).
3.2		
4.1		
4.2	Read **PC**	Store as Operand.
5.1		Inc. **PC**. If !Jump, end instruction (next half-cycle is 7.2).
5.2		
6.1		Address = **PC** + i8(Operand).
		PC.L = Address.L.
6.2		
7.1		**PC**.H = Address.H.
7.2	Read **PC**	Store as OpCode.
+X.1		

AND (24)	Cycles: 3	Size: 2
	Direct Page d (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Operand.
2.1		Inc. **PC**. Address = (Operand | ((**PSW** & **P** flag) << 3)).
2.2	Read Address	Store as Operand.
3.1		**A** &= Operand. **N** flag = (**A** & $80), **Z** flag = !**A**.
3.2	Read **PC**	Store as OpCode.
+X.1		

AND (25)	Cycles: 4	Size: 3
	Absolute !a (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Address.L.
2.1		Inc. **PC**.
2.2	Read **PC**	Store as Address.H.
3.1		Inc. **PC**.
3.2	Read Address	Store as Operand.
4.1		**A** &= Operand. **N** flag = (**A** & $80), **Z** flag = !**A**.
4.2	Read **PC**	Store as OpCode.
+X.1		

AND (26)	Cycles: 3	Size: 1
	Indirect X (X) (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2		
2.1		
2.2	Read (X | ((**PSW** & **P** flag) << 3))	Store as Operand.
3.1		**A** &= Operand. **N** flag = (**A** & $80), **Z** flag = !**A**.
3.2	Read **PC**	Store as OpCode.
+X.1		

AND (27)	Cycles: 6	Size: 2
	X-Indexed Indirect [d+X] (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Operand.
2.1		Inc. **PC**.
2.2		
3.1		Address = ((**X** + Operand) & $FF) | ((**PSW** & **P** flag) << 3).
3.2	Read Address	Store as Pointer.L.
4.1		
4.2	Read ((Address + 1) & $FF) | ((**PSW** & **P** flag) << 3)	Store as Pointer.H.
5.1		
5.2	Read Pointer	Store as Operand.
6.1		**A** &= Operand. **N** flag = (**A** & $80), **Z** flag = !**A**.
6.2	Read **PC**	Store as OpCode.
+X.1		

AND (28)	Cycles: 2	Size: 2
	Immediate	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Operand.
2.1		Inc. **PC**. **A** &= Operand. **N** flag = (**A** & $80), **Z** flag = !**A**.
2.2	Read **PC**	Store as OpCode.
+X.1		

AND (29)	Cycles: 6	Size: 3
	Direct Page to Direct Page dd, ds (Read/Modify/Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Operand.
2.1		Inc. **PC**. Address = (Operand | ((**PSW** & **P** flag) << 3)).
2.2	Read Address	Store as Operand0.
3.1		
3.2	Read **PC**	Store as Operand.
4.1		Inc. **PC**. Address = (Operand | ((**PSW** & **P** flag) << 3)).
4.2	Read Address	Store as Operand1.
5.1		Operand0 &= Operand1. **N** flag = (Operand0 & $80), **Z** flag = !Operand0.
5.2	Write to Address	Write Operand0.
6.1		
6.2	Read **PC**	Store as OpCode.
+X.1		

OR1 (2A)	Cycles: 5	Size: 3
	Absolute Boolean Bit /m.b	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Address.L.
2.1		Inc. **PC**.
2.2	Read **PC**	Store as Address.H.
3.1		Inc. **PC**.
3.2	Read (Address & $1FFF)	Store as Operand.
4.1		Bit = Address >> 13.
		C flag |= ((Operand & (1 << Bit)) == 0).
4.2		
5.1		
5.2	Read **PC**	Store as OpCode.
+X.1		

ROL (2B)	Cycles: 4	Size: 2
	Direct Page d (Read/Modify/Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Operand.
2.1		Inc. **PC**. Address = (Operand | ((**PSW** & **P** flag) << 3)).
2.2	Read Address	Store as Operand.
3.1		Tmp = **C** flag. **C** flag = (Operand & $80). Operand = (Operand << 1) | Tmp. **N** flag = (Operand & $80), **Z** flag = !Operand.
3.2	Write to Address	Write Operand.
4.1		
4.2	Read **PC**	Store as OpCode.
+X.1		

ROL (2C)	Cycles: 5	Size: 3
	Absolute !a (Read/Modify/Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Address.L.
2.1		Inc. **PC**.
2.2	Read **PC**	Store as Address.H.
3.1		Inc. **PC**.
3.2	Read Address	Store as Operand.
4.1		Tmp = **C** flag. **C** flag = (Operand & $80). Operand = (Operand << 1) | Tmp. **N** flag = (Operand & $80), **Z** flag = !Operand.
4.2	Write to Address	Write Operand.
5.1		
5.2	Read **PC**	Store as OpCode.
+X.1		

PUSH (2D)	Cycles: 4	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Discard.
2.1		
2.2	Write to u8(**SP**+0) | $0100	Push **A** onto stack.
3.1		Dec. **SP** by 1.
3.2		
4.1		
4.2	Read **PC**	Store as OpCode.
+X.1		

CBNE (2E)	Cycles: 5-7	Size: 3
	Direct Page Relative d, r	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Operand.
2.1		Inc. **PC**. Address = (Operand | ((**PSW** & **P** flag) << 3)).
2.2	Read Address	Store as Operand.
3.1		Set Jump to (**A** != Operand).
3.2		
4.1		
4.2	Read **PC**	Store as Operand.
5.1		Inc. **PC**. If !Jump, end instruction (next half-cycle is 7.2).
5.2		
6.1		Address = **PC** + i8(Operand). **PC**.L = Address.L.
6.2		
7.1		**PC**.H = Address.H.
7.2	Read **PC**	Store as OpCode.
+X.1		

BRA (2F)	Cycles: 4	Size: 2
	Relative r	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**. Set Jump to true.
1.2	Read **PC**	Store as Operand.
2.1		Inc. **PC**. If !Jump, end instruction (next half-cycle is 4.2).
2.2		
3.1		Address = **PC** + i8(Operand). **PC**.L = Address.L.
3.2		
4.1		**PC**.H = Address.H.
4.2	Read **PC**	Store as OpCode.
+X.1		

BMI (30)	Cycles: 2-4	Size: 2
	Relative r	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**. Set Jump to (**PSW** & $80) == $80.
1.2	Read **PC**	Store as Operand.
2.1		Inc. **PC**. If !Jump, end instruction (next half-cycle is 4.2).
2.2		
3.1		Address = **PC** + i8(Operand). **PC**.L = Address.L.
3.2		
4.1		**PC**.H = Address.H.
4.2	Read **PC**	Store as OpCode.
+X.1		

TCALL3 (31)	Cycles: 8	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Operand.
2.1		
2.2		
3.1		
3.2	Write to u8(**SP**+0) | $0100	Push **PC**.H onto stack.
4.1		
4.2	Write to u8(**SP**-1) | $0100	Push **PC**.L onto stack.
5.1		Dec. **SP** by 2.
5.2		
6.1		
6.2	Read $FFD8	Store as **PC**.L.
7.1		
7.2	Read $FFD9	Store as **PC**.H.
8.1		
8.2	Read **PC**	Store as OpCode.
+X.1		

CLR1 (32)	Cycles: 4	Size: 2
	Direct Page Bit d.b (Read/Modify/Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Operand.
2.1		Inc. **PC**. Address = (Operand | ((**PSW** & **P** flag) << 3)).
2.2	Read Address	Store as Operand.
3.1		Operand = (Operand & ~$02) | $00.
3.2	Write to Address	Write Operand.
4.1		
4.2	Read **PC**	Store as OpCode.
+X.1		

BBC1 (33)	Cycles: 5-7	Size: 3
	Direct Page Bit Relative d.b, r	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Operand.
2.1		Inc. **PC**. Address = (Operand | ((**PSW** & **P** flag) << 3)).
2.2	Read Address	Store as Operand.
3.1		Set Jump to (Operand & (1 << 1)) == (0 << 1).
3.2		
4.1		
4.2	Read **PC**	Store as Operand.
5.1		Inc. **PC**. If !Jump, end instruction (next half-cycle is 7.2).
5.2		
6.1		Address = **PC** + i8(Operand).
		PC.L = Address.L.
6.2		
7.1		**PC**.H = Address.H.
7.2	Read **PC**	Store as OpCode.
+X.1		

AND (34)	Cycles: 4	Size: 2
	X-Indexed Direct Page d+X (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Operand.
2.1		Inc. **PC**.
2.2		
3.1		Address = ((**X** + Operand) & $FF) | ((**PSW** & **P** flag) << 3).
3.2	Read Address	Store as Operand.
4.1		**A** &= Operand. **N** flag = (**A** & $80), **Z** flag = !**A**.
4.2	Read **PC**	Store as OpCode.
+X.1		

AND (35)	Cycles: 5	Size: 3
	X-Indexed Absolute !a+X (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Address.L.
2.1		Inc. **PC**.
2.2	Read **PC**	Store as Address.H.
3.1		Inc. **PC**.
3.2		
4.1		Pointer = (**X** + Address).
4.2	Read Pointer	Store as Operand.
5.1		**A** &= Operand. **N** flag = (**A** & $80), **Z** flag = !**A**.
5.2	Read **PC**	Store as OpCode.
+X.1		

AND (36)	Cycles: 5	Size: 3
	Y-Indexed Absolute !a+Y (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Address.L.
2.1		Inc. **PC**.
2.2	Read **PC**	Store as Address.H.
3.1		Inc. **PC**.
3.2		
4.1		Pointer = (**Y** + Address).
4.2	Read Pointer	Store as Operand.
5.1		**A** &= Operand. **N** flag = (**A** & $80), **Z** flag = !**A**.
5.2	Read **PC**	Store as OpCode.
+X.1		

AND (37)	Cycles: 6	Size: 2
	Indirect Y-Indexed [d]+Y (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Operand.
2.1		Inc. **PC**.
2.2		
3.1		Address = (Operand | ((**PSW** & **P** flag) << 3)).
3.2	Read Address	Store as Pointer.L.
4.1		
4.2	Read ((Address + 1) & $FF) | ((**PSW** & **P** flag) << 3)	Store as Pointer.H.
5.1		Address = (**Y** + Pointer).
5.2	Read Address	Store as Operand.
6.1		**A** &= Operand. **N** flag = (**A** & $80), **Z** flag = !A.
6.2	Read **PC**	Store as OpCode.
+X.1		

AND (38)	Cycles: 5	Size: 3
	Immediate Data to Direct Page d, #i (Read/Modify/Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Operand0.
2.1		Inc. **PC**.
2.2	Read **PC**	Store as Operand.
3.1		Inc. **PC**. Address = (Operand | ((**PSW** & **P** flag) << 3)).
3.2	Read Address	Store as Operand1.
4.1		Operand0 &= Operand1. **N** flag = (Operand0 & $80), **Z** flag = !Operand0.
4.2	Write to Address	Write Operand0.
5.1		
5.2	Read **PC**	Store as OpCode.
+X.1		

AND (39)	Cycles: 5	Size: 1
	Indirect Page to Indirect Page (X), (Y) (Read/Modify/Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Discard.
2.1		
2.2	Read (**Y** | ((**PSW** & **P** flag) << 3))	Store as Operand0.
3.1		
3.2	Read (**X** | ((**PSW** & **P** flag) << 3))	Store as Operand1.
4.1		Operand0 &= Operand1. **N** flag = (Operand0 & $80), **Z** flag = !Operand0.
4.2	Write to (**X** | ((**PSW** & **P** flag) << 3))	Write Operand0.
5.1		
5.2	Read **PC**	Store as OpCode.
+X.1		

INCW (3A)	Cycles: 6	Size: 2
	Direct Page d (Read/Modify/Write 16)	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Operand.
2.1		Inc. **PC**. Address = (Operand | ((**PSW** & **P** flag) << 3)).
2.2	Read Address	Store as Operand.
3.1		Overflow = (Operand == $FF). Operand += 1. LowZero = (Operand == 0).
3.2	Write to Address	Write Operand.
4.1		
4.2	Read ((Address + 1) & $FF) | ((**PSW** & **P** flag) << 3)	Store as Operand.
5.1		Operand += Overflow. **N** flag = (Operand & $80), **Z** flag = (LowZero && !Operand).
5.2	Write to ((Address + 1) & $FF) | ((**PSW** & **P** flag) << 3)	Write Operand.
6.1		
6.2	Read **PC**	Store as OpCode.
+X.1		

ROL (3B)	Cycles: 5	Size: 2
	X-Indexed Direct Page d+X (Read/Modify/Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Operand.
2.1		Inc. **PC**.
2.2		
3.1		Address = ((**X** + Operand) & $FF) | ((**PSW** & **P** flag) << 3).
3.2	Read Address	Store as Operand.
4.1		Tmp = **C** flag. **C** flag = (Operand & $80). Operand = (Operand << 1) | Tmp. **N** flag = (Operand & $80), **Z** flag = !Operand.
4.2	Write to Address	Write Operand.
5.1		
5.2	Read **PC**	Store as OpCode.
+X.1		

ROL (3C)	Cycles: 2	Size: 1
	Accumulator A	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Discard.
2.1		Tmp = **C** flag. **C** flag = (**A** & $80). **A** = (**A** << 1) | Tmp. **N** flag = (**A** & $80), **Z** flag = !**A**.
2.2	Read **PC**	Store as OpCode.
+X.1		

INC (3D)	Cycles: 2	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2		
2.1		**X** += 1. **N** flag = (**X** & $80), **Z** flag = !**X**.
2.2	Read **PC**	Store as OpCode.
+X.1		

CMP (3E)	Cycles: 3	Size: 2
	Direct Page d (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Operand.
2.1		Inc. **PC**. Address = (Operand | ((**PSW** & **P** flag) << 3)).
2.2	Read Address	Store as Operand.
3.1		Tmp = **X** - Operand. **C** flag = (Tmp >= 0), **N** flag = (Tmp & $80), **Z** = !Tmp.
3.2	Read **PC**	Store as OpCode.
+X.1		

CALL (3F)	Cycles: 8	Size: 3
	Absolute Call !a	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Address.L.
2.1		Inc. **PC**.
2.2	Read **PC**	Store as Address.H.
3.1		Inc. **PC**.
3.2		
4.1		
4.2	Write to u8(**SP**+0) | $0100	Push **PC**.H onto stack.
5.1		
5.2	Write to u8(**SP**-1) | $0100	Push **PC**.L onto stack.
6.1		Dec. **SP** by 2.
6.2		
7.1		**PC**.L = Address.L.
7.2		
8.1		**PC**.H = Address.H.
8.2	Read **PC**	Store as OpCode.
+X.1		

SETP (40)	Cycles: 2	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2		
2.1		**PSW** = (**PSW** | $20).
2.2	Read **PC**	Store as OpCode.
+X.1		

TCALL4 (41)	Cycles: 8	Size: 1
	Implied	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Operand.
2.1		
2.2		
3.1		
3.2	Write to u8(**SP**+0) | $0100	Push **PC**.H onto stack.
4.1		
4.2	Write to u8(**SP**-1) | $0100	Push **PC**.L onto stack.
5.1		Dec. **SP** by 2.
5.2		
6.1		
6.2	Read $FFD6	Store as **PC**.L.
7.1		
7.2	Read $FFD7	Store as **PC**.H.
8.1		
8.2	Read **PC**	Store as OpCode.
+X.1		

SET1 (42)	Cycles: 4	Size: 2
	Direct Page Bit d.b (Read/Modify/Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Operand.
2.1		Inc. **PC**. Address = (Operand | ((**PSW** & **P** flag) << 3)).
2.2	Read Address	Store as Operand.
3.1		Operand = (Operand & ~$04) | $04.
3.2	Write to Address	Write Operand.
4.1		
4.2	Read **PC**	Store as OpCode.
+X.1		

BBS2 (43)	Cycles: 5-7	Size: 3
	Direct Page Bit Relative d.b, r	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Operand.
2.1		Inc. **PC**. Address = (Operand | ((**PSW** & **P** flag) << 3)).
2.2	Read Address	Store as Operand.
3.1		Set Jump to (Operand & (1 << 2)) == (1 << 2).
3.2		
4.1		
4.2	Read **PC**	Store as Operand.
5.1		Inc. **PC**. If !Jump, end instruction (next half-cycle is 7.2).
5.2		
6.1		Address = **PC** + i8(Operand).
		PC.L = Address.L.
6.2		
7.1		**PC**.H = Address.H.
7.2	Read **PC**	Store as OpCode.
+X.1		

EOR (44)	Cycles: 3	Size: 2
	Direct Page d (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Operand.
2.1		Inc. **PC**. Address = (Operand | ((**PSW** & **P** flag) << 3)).
2.2	Read Address	Store as Operand.
3.1		**A** ^= Operand. **N** flag = (**A** & $80), **Z** flag = !**A**.
3.2	Read **PC**	Store as OpCode.
+X.1		

EOR (45)	Cycles: 4	Size: 3
	Absolute !a (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Address.L.
2.1		Inc. **PC**.
2.2	Read **PC**	Store as Address.H.
3.1		Inc. **PC**.
3.2	Read Address	Store as Operand.
4.1		**A** ^= Operand. **N** flag = (**A** & $80), **Z** flag = !**A**.
4.2	Read **PC**	Store as OpCode.
+X.1		

EOR (46)	Cycles: 3	Size: 1
	Indirect X (X) (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2		
2.1		
2.2	Read (**X** | ((**PSW** & **P** flag) << 3))	Store as Operand.
3.1		**A** ^= Operand. **N** flag = (**A** & $80), **Z** flag = !**A**.
3.2	Read **PC**	Store as OpCode.
+X.1		

EOR (47)	Cycles: 6	Size: 2
	X-Indexed Indirect [d+X] (Read)	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Operand.
2.1		Inc. **PC**.
2.2		
3.1		Address = ((**X** + Operand) & $FF) | ((**PSW** & **P** flag) << 3).
3.2	Read Address	Store as Pointer.L.
4.1		
4.2	Read ((Address + 1) & $FF) | ((**PSW** & **P** flag) << 3)	Store as Pointer.H.
5.1		
5.2	Read Pointer	Store as Operand.
6.1		**A** ^= Operand. **N** flag = (**A** & $80), **Z** flag = !**A**.
6.2	Read **PC**	Store as OpCode.
+X.1		

EOR (48)	Cycles: 2	Size: 2
	Immediate	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Operand.
2.1		Inc. **PC**. **A** ^= Operand. **N** flag = (**A** & $80), **Z** flag = !**A**.
2.2	Read **PC**	Store as OpCode.
+X.1		

EOR (49)	Cycles: 6	Size: 3
	Direct Page to Direct Page dd, ds (Read/Modify/Write)	
Cycle	R/W	Desc.
-X.2	Read **PC**	Store as OpCode.
1.1		Inc. **PC**.
1.2	Read **PC**	Store as Operand.
2.1		Inc. **PC**. Address = (Operand | ((**PSW** & **P** flag) << 3)).
2.2	Read Address	Store as Operand0.
3.1		
3.2	Read **PC**	Store as Operand.
4.1		Inc. **PC**. Address = (Operand | ((**PSW** & **P** flag) << 3)).
4.2	Read Address	Store as Operand1.
5.1		Operand0 ^= Operand1. **N** flag = (Operand0 & $80), **Z** flag = !Operand0.
5.2	Write to Address	Write Operand0.
6.1		
6.2	Read **PC**	Store as OpCode.
+X.1		